Md. Taifur Hossain.

SOFTWARE ENGINEERING INTERN

+880 1864 245617
taifurrahman0606@gmail.com
Dhaka, Bangladesh
github.com/Daceyyreal · daceyyreal.github.io
linkedin.com/in/taifur-hossain-siam-b67447433

Final-year Computer Science undergraduate who ships working software. Built and released **four open-source tools** across 3D vision, model compression, and developer tooling — each with tests, CI, documentation, and a tagged release — plus a **first-author paper accepted at ICCA 2026 (ACM)**. Comfortable across C, C++, C#, Python, and JavaScript, and used to owning a problem end to end, from prototype to a documented release others can run. Open to software engineering internships across backend, full-stack, systems, and ML.

TECHNICAL SKILLS

LANGUAGES	C, C++, C#, Python, JavaScript / TypeScript, Bash
ML & VISION	PyTorch, NumPy, 3D Gaussian Splatting, NeRF, Nerfstudio, gsplat, COLMAP, model quantization, RAG & embeddings (sentence-transformers)
WEB & UI	Three.js, Vite, Streamlit, HTML / CSS
TOOLS & INFRA	Git, Docker, GitHub Actions (CI/CD), GHCR, Linux, pytest, ruff

PROJECTS

splat-slim — GitHub

Training-free compression tool for trained 3D Gaussian Splatting models.

- Compresses trained 3DGS models by **~74% with under 3 dB quality loss**, with no retraining.
- Built a four-stage pipeline: adaptive opacity pruning, outlier clipping, spherical-harmonic reduction, and quantization.
- Designed a self-decodable mixed-precision format storing per-group ranges in a sidecar `.meta.npz`, so compressed files still open in standard viewers.
- Shipped as both a CLI and a library, with automated tests and linting on every push and a tagged `v0.1.0` release.

Tech Stack: Python, Typer, NumPy, plyfile, pytest, ruff, GitHub Actions.

nerf-prep — GitHub

One-command Dockerized pipeline turning a folder of photos into a training-ready 3D dataset.

- Packaged a fragile multi-tool chain (COLMAP to Nerfstudio) into **a single reproducible Docker command**, published to GHCR automatically by CI.
- Automatic matcher selection by capture type, plus GPU/CPU auto-detection with a CPU fallback so the same command runs on a cloud GPU or a laptop.
- Validates input before a long run — flags double extensions, too-few images, and stray files — and prints a registration-rate summary at the end.
- Verified end to end on a real 40-photo capture: **40 of 40 images registered**.

Tech Stack: Python, Typer, Docker, COLMAP, Nerfstudio, GitHub Actions, GHCR.

splat-lab — Live Demo · GitHub

Browser-based, real-time demo of the compression pipeline.

- Load a 3D scene and watch file size and visual quality trade off **live on the GPU** while dragging sliders.
- Ported the Python compression maths to TypeScript so the browser reproduces the same results.
- Runs entirely client-side with no backend, and deploys automatically to GitHub Pages.

Tech Stack: TypeScript, Three.js, Vite, GitHub Pages.

paper-chat — GitHub

Retrieval-augmented question answering over a PDF library, with citations you can verify.

- **Every answer cites the exact paper and page** it came from, and the app declines to answer when the sources do not cover the question.
- Parses PDFs per page so each retrieved chunk carries a (paper, page) tag, enabling cross-paper comparison questions.
- Swappable LLM backend (Groq or Gemini) behind a small interface, selectable by environment variable.

- Retrieval and citation logic unit-tested with fake models, so the test suite runs with no API key and no model download.

Tech Stack: Python, Streamlit, sentence-transformers, PyMuPDF, NumPy.

PUBLICATION

Beyond Naive INT8: Adaptive Post-Training Compression for 3D Gaussian Splatting · ACCEPTED

M. T. Hossain, S. S. Sourav, M. A. Hossain. Proc. ICCA 2026 (ACM), Dhaka.

- First author. Built the full experimental pipeline and evaluated it across three scenes on a single 16 GB GPU.
- Achieved a **74% size reduction without retraining**, and diagnosed a quantization failure mode that a sub-group range scheme recovers to within 0.1–0.7 dB.

EDUCATION

International University of Business Agriculture & Technology (IUBAT) Dhaka, BD

B.Sc. in Computer Science & Engineering (BCSE) Expected 2027

CGPA 3.4 / 4.00

Higher Secondary Certificate (HSC) Dhaka, BD · 2019 – 2021

GPA 5.00 / 5.00

Secondary School Certificate (SSC) Dhaka, BD · 2018

GPA 5.00 / 5.00

LANGUAGES

ENGLISH	Proficient
BANGLA	Native